

LENA LIEBICH

HARVARD BUSINESS SCHOOL AI INSTITUTE, DIGITAL RESKILLING LAB

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Academic Appointments

Harvard Business School AI Institute

Postdoctoral Fellow, Digital Reskilling Lab

Boston, MA, US

2026 – present

Leibniz Institute for Financial Research SAFE

Research Affiliate

Frankfurt am Main, Germany

2026 – present

Research Interests

Human-AI collaboration, Technology adoption, Behavioral economics

Publications

[1] Knowing (Not) to Know: Explainable Artificial Intelligence and Human Metacognition

with Moritz von Zahn, Ekaterina Jussupow, Oliver Hinz and Kevin Bauer (2025)

Forthcoming in Information Systems Research, published online in Articles in Advance.

Abstract: The use of explainable artificial intelligence (XAI) to render black-box artificial intelligence (AI) more interpretable to humans is gaining practical relevance. Prior research has shown that XAI can influence how humans “think”. Yet, little is known about whether XAI also affects how people “think about their thinking” (i.e., their metacognitive processes). We address this gap by investigating whether XAI affects metacognitive processes that guide human confidence judgments about their ability to perform a task and thereby, their decision whether to delegate the task to AI. We conducted two incentivized experiments in which domain experts repeatedly performed prediction tasks, with the option to delegate each task to an AI. We exogenously varied whether participants initially received explanations about AI’s overall prediction logic. We find that AI explanations improve how well humans understand their own ability to perform the task (metacognitive accuracy). This improvement causally increases both the frequency and effectiveness of human-to-AI delegation. Additional analyses show that these effects primarily occur when explanations reveal to humans that AI’s prediction logic diverges from their own, leading to a systematic reduction of overconfidence. Our findings highlight metacognitive processes as a central, previously overlooked channel through which AI explanations can influence human-AI collaboration. We discuss practical implications of our results for organizations implementing XAI to comply with regulatory transparency requirements as, for example, outlined in the European Union Artificial Intelligence Act.

PAPER

DOI

[2] Can GPT Mimic Human Preferences? An Empirical and Structural Investigation

with Kevin Bauer and Michael Kosfeld (2025)

Proceedings of the European Conference on Information Systems (ECIS 2025)

Abstract: We examine the depth of large language models’ (LLMs) potential to serve as tools for simulating human decision-making processes. Using GPT-4 Turbo in a sequential prisoner’s dilemma, we find significant discrepancies between LLM-generated and human behaviors: GPT consistently exhibits a higher propensity to cooperate and forms overly optimistic expectations of human cooperation, leading to outcomes that diverge from observed human decision-making patterns. Yet, GPT’s decisions do not appear random; they align with a formal model of human preferences for fairness and efficiency that explains the behavior of most human participants in similar settings. We find a striking

contrast: while GPT may fall short as accurate surrogate for human decisions at the surface level, the model nonetheless exhibits a capacity to reflect the underlying preferences that drive human behavior in strategic contexts. Our findings have implications for LLMs' potential to serve as a managerial or research tool in simulating human behaviors.

PAPER

Work in Progress

[3] Responsive Advisors, Indifferent Clients: Evidence from a Field Experiment on the Role of Explainable AI in Financial Advice (Working Title)

First-authored, with Kevin Bauer, Cathy Yang, Xitong Li and Oliver Hinz (2026)

[4] Personalized Communication and Repayment (Working Title)

with Florian Hett, Andrej Gill, Christine Laudenbach and Stephan Siegel (2026)

Policy Work (Pre-PhD)

[5] Unconventionally Green: Monetary Policy between Engagement and Conflicting Goals

with Lukas Nöh, Felix Rutkowski and Milena Schwarz (2023)

Review of Economics, 74(1), 53–77.

PAPER

DOI

[6] Current Developments in Green Finance

with Lukas Nöh, Felix Rutkowski and Milena Schwarz (2020)

German Council of Economic Experts, Working Paper 05/2020.

PAPER

Education

Goethe University Frankfurt (GSEFM)

Frankfurt am Main, Germany

PhD in Economics, *summa cum laude*

2020 – 2026

Dissertation: *Essays on Human-AI Interaction and Household Finance*

Committee: Christine Laudenbach (first supervisor), Michael Kosfeld (second supervisor), Guido Friebel, Florian Hett

University of Melbourne

Melbourne, Australia

Visiting Scholar, Centre for Brain, Mind and Markets

2025 – 2026

Two-month research stay hosted by Carsten Murawski

CUHK Business School

Hong Kong SAR

Visiting Scholar, Department of Decisions, Operations and Technology

2025

Two-month research stay hosted by Jenny (Qianran) Jin

Goethe University Frankfurt (GSEFM)

Frankfurt am Main, Germany

MSc in Quantitative Economics

2020 – 2025

Electives: Applied Econometrics, Household Finance, Experimental Finance

University of Bonn

Bonn, Germany

BSc in Economics

2016 – 2020

Electives: Experimental Economics, Game Theory, Contract Theory, Applied Microeconometrics

Teaching

Gutenberg School of Business, JGU Mainz

Teaching Assistant, *Machine Learning for Management Decisions*
MBA course held by Kevin Bauer

Mainz, Germany
2025

University of Bonn

Teaching Assistant, *Financial Markets and Institutions*
BSc Economics held by Isabel Schnabel

Bonn, Germany
2019

Awards, Grants & Scholarships

Publication Prize for Early Career Researchers

For the forthcoming publication of *Knowing (Not) to Know: Explainable Artificial Intelligence and Human Metacognition in Information Systems Research*

2026

Award for Excellence in Teaching

Joint with Kevin Bauer, for the MBA course *Machine Learning for Management Decisions*, Gutenberg School of Business, JGU Mainz

2026

SAFE Young Researcher Travel Grant

Fully-funded research visits to CUHK Business School and the University of Melbourne

2025

Best Discussion Award

9th SAFE Household Finance Workshop, sponsored by Behavioural Finance e.V. (€1,000)

2025

First-Year PhD Scholarship

Leibniz Institute for Financial Research SAFE

2020 – 2021

Conferences, Seminars and Invited Talks

- 4th Workshop on Behavioral, Digital and Financial Economics, JGU Mainz (presentation) 2026
- Digital Economy Workshop (presentation)
- AI Plus Management Doctoral Consortium, UCL School of Management (presentation, cancelled)
- SAFE Brown Bag Seminar (presentation)
- SAFE Junior Scholars Workshop: The New World of Corporate Boards (presentation) 2025
- 3rd Workshop on Behavioral, Digital and Financial Economics, JGU Mainz (presentation)
- University of Oxford, Faculty of Law (presentation)
- 9th SAFE Household Finance Workshop (discussion)
- CUHK Business School (presentation)
- 38th Australasian Finance and Banking Conference, UNSW Business School (presentation)
- 2nd Workshop on Behavioral, Digital and Financial Economics, JGU Mainz (presentation) 2024
- SAFE Brown Bag Seminar (presentation)
- 9th Spring School in Behavioral Economics, UCSD Rady School of Management (poster)
- Machine+Behavior Conference, MPI / Toulouse School of Economics / University of British Columbia (poster)
- 8th SAFE Household Finance Workshop (discussion)
- EconTribute Workshop on Human-AI Collaboration, University of Cologne (lightning talk)

- 1st Workshop on Behavioral, Digital and Financial Economics, JGU Mainz (presentation) 2023
- PhD course *Subjective Beliefs, Attention and Economic Behavior*, University of Copenhagen (poster)
- 1st Bonn-Frankfurt-Mannheim PhD Conference (presentation)
- Workshop on Interactions of Humans and Algorithms, TU Berlin (poster)
- SAFE Brown Bag Seminar (presentation)
- Society for Experimental Finance Summer School, University of Sofia
- 7th SAFE Household Finance Workshop (presentation)
- CORA Conference on Robots and Automation, Goethe University (presentation)
- ZEW / University of Mannheim Experimental Seminar (presentation)
- 6th SAFE Household Finance Workshop (discussion) 2022

Academic Service

- Supervision of Master's and Bachelor's theses (Leibniz Institute for Financial Research SAFE) 2022 – 2026
- Representative of PhD Researchers (Leibniz Institute for Financial Research SAFE) 2022 – 2024
- Member of Student Representatives (GSEFM, Goethe University Frankfurt) 2020 – 2024
- Organizing Committee, 1st Bonn-Frankfurt-Mannheim PhD Conference with keynote by Ulrike Malmendier (Goethe University Frankfurt) 2023
- Referee** 2025
- European Conference on Information Systems (ECIS)

Professional Experience

Academic

- Leibniz Institute for Financial Research SAFE** Frankfurt am Main, Germany
Junior Researcher / Doctoral Student, Department of Household Finance 2022 – 2025
- European Central Bank** Frankfurt am Main, Germany
Academic Researcher, DG Macroeconomic Policy and Financial Stability 2022 – 2023
- European Central Bank** Frankfurt am Main, Germany
Trainee, DG Macroeconomic Policy and Financial Stability 2022
- Leibniz Institute for Financial Research SAFE** Frankfurt am Main, Germany
Student Research Assistant 2021 – 2022
- German Council of Economic Experts (GCEE)** Wiesbaden, Germany
Research Intern, Scientific Staff 2020

Non-Academic

- Consulting, media and language tutoring** 2015 – 2020
- Consulting Intern and Assistant (organizational change management), CPC Unternehmensmanagement AG, Bonn & Frankfurt am Main, Germany
- Student Assistant, Editorial Office *Documentaries*, phoenix / ZDF, Bonn, Germany
- German Language Tutor (Gap Year), Bilbao, Spain

Affiliations

- Harvard Business School AI Institute
- Leibniz Institute for Financial Research SAFE (Research Affiliate)
- Behavioural Finance e.V. (Member)
- Mentoring Hessen, ProCareer.Doc Program (2024 cohort)

Personal

Software	Stata, Python, LaTeX, Git, oTree, MS Office
Languages	German (native), English (fluent), Spanish (fluent)
Citizenship	German

References

Available on request.